

Summary Table (Comparative Analysis of References Cited in the Selected Paper)

Ref. No.	Year	AI/ML Technique	Healthcare Application	Security / Privacy Method	Key Contribution	Research Gap / Limitation
Ref. 28	2023	Blockchain-enabled Federated Learning	Internet of Medical Things (IoMT)	Blockchain	Improved trust, transparency, and secure data sharing	High computational and storage overhead
Ref. 29	2023	Secure Federated Learning	IoMT Healthcare	Secure Aggregation	Protected distributed model training without exposing patient data	Increased communication cost
Ref. 32	2023	Federated Learning	Medical Applications	Privacy-Preserving Learning	Enabled collaborative disease prediction while preserving data privacy	Performance depends on data heterogeneity
Ref. 33	2023	Federated Learning Survey	Healthcare Informatics	Differential Privacy	Comprehensive overview of FL techniques	Limited discussion on adaptive security threats
Ref. 34	2024	Federated Learning	Smart Healthcare	Differential Privacy + Secure Aggregation	Reviewed privacy-preserving AI solutions	Accuracy–privacy trade-off remains unresolved

Ref. 35	2024	Federated Learning	Heart Disease Prediction	Secure Model Aggregation	Improved prediction accuracy	Requires large-scale collaborative datasets
Ref. 36	2024	Federated Learning	Disease Prediction	Privacy-Preserving Analytics	Enabled multi-hospital collaborative learning	Vulnerable to Byzantine model poisoning attacks
Ref. 37	2024	AI + Federated Learning	Smart Healthcare	Secure Federated Learning	Enhanced intelligent healthcare services	Limited scalability in heterogeneous IoMT
Ref. 112	2025	Federated Learning Survey	Healthcare, Finance & IoT	Privacy-Preserving FL	Summarized recent FL applications	Lack of practical deployment strategies
Ref. 113	2025	Federated Learning	Healthcare Systems	Differential Privacy	Identified security challenges	Communication efficiency remains a challenge
Ref. 115	2025	Federated Learning	Internet of Things	Secure Aggregation	Reviewed FL architectures	High resource consumption for edge devices
Ref. 118	2025	Explainable AI	AI-driven Healthcare	Ethical AI Framework	Addressed fairness and transparency	Limited implementation in real-world clinical environments

Overall Comparative Analysis:

The reviewed studies demonstrate that Federated Learning has become a promising solution for privacy-preserving healthcare by enabling collaborative model training without sharing sensitive patient data. Recent research integrates advanced techniques such as Blockchain, Differential Privacy, Secure Aggregation, and Explainable AI to improve security, transparency, and model reliability. However, most existing approaches still face significant challenges, including communication overhead, computational complexity, data heterogeneity, scalability issues, and the trade-off between privacy and model accuracy. These limitations indicate that more efficient, secure, and scalable Federated Learning frameworks are required for real-world AI-driven healthcare applications.